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## Department of Engineering

Session 2026-27

**PROMPT ENGINEERING**

**(Course Code: BCO610A)**

# HW-05: THE HONEST REPOSITORY

GitHub & AI-Generated Documentation


Submitted By:

Name –Aditya Ranjan

Enrollment No. – 25BCON1294

Branch – B.Tech. CSE (AI&DS)

Semester – 3

Submitted To:

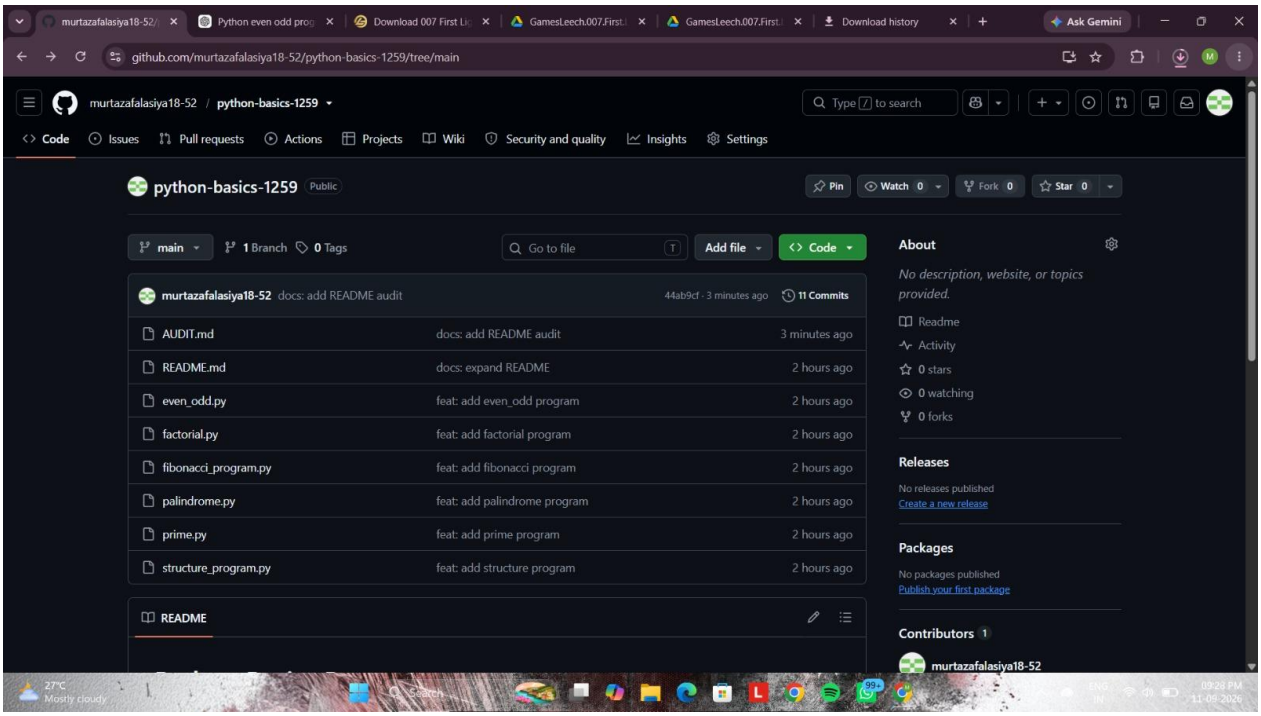
Mr. Ajay Sharma

Prompt Engineering

# 1. Public Repository + Repository Hygiene

## Repository Details

Repository name: python-basics-25BCON1259
Visibility: Public
Repository link: https://github.com/murtazafalasiya18-52/python-basics-1259



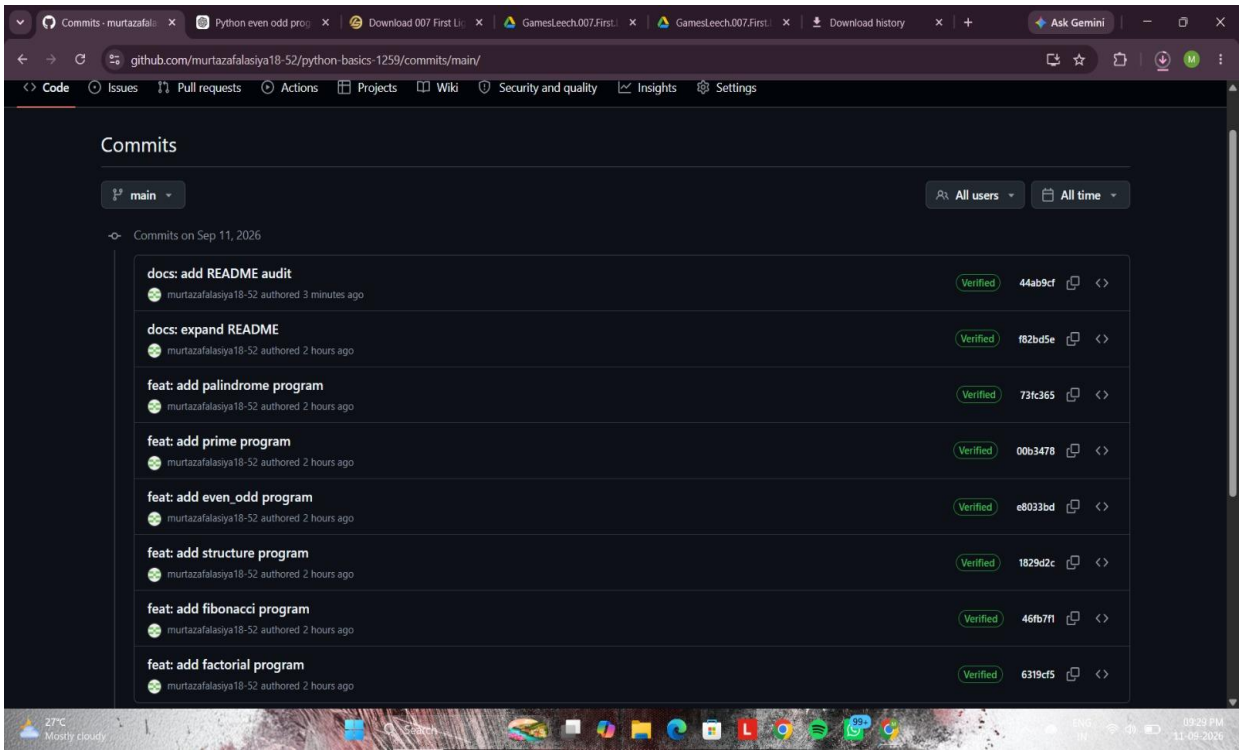

## Six Programs

| No. | Program / File | Purpose |
|---|---|---|
| 1 | factorial.py | Calculate factorial of a given number. |
| 2 | fibonacci.py | Generate a Fibonacci series. |
| 3 | structure.py | Store and display student information. |
| 4 | even_odd.py | Check whether a number is even or odd. |
| 5 | prime.py | Check whether a number is prime. |
| 6 | palindrome.py | Check whether a string/number is a palindrome. |

# 2. Commit History + AI Message Comparison

## My Commit Messages

| Commit | My message | Change represented |
|---|---|---|
| 1 | feat: add factorial program | Added factorial.py |
| 2 | feat: add fibonacci program | Added fibonacci.py |
| 3 | feat: add student structure | Added structure.py |
| 4 | feat: add even odd checker | Added even_odd.py |
| 5 | feat: add prime checker | Added prime.py |
| 6 | feat: add palindrome checker | Added palindrome.py |

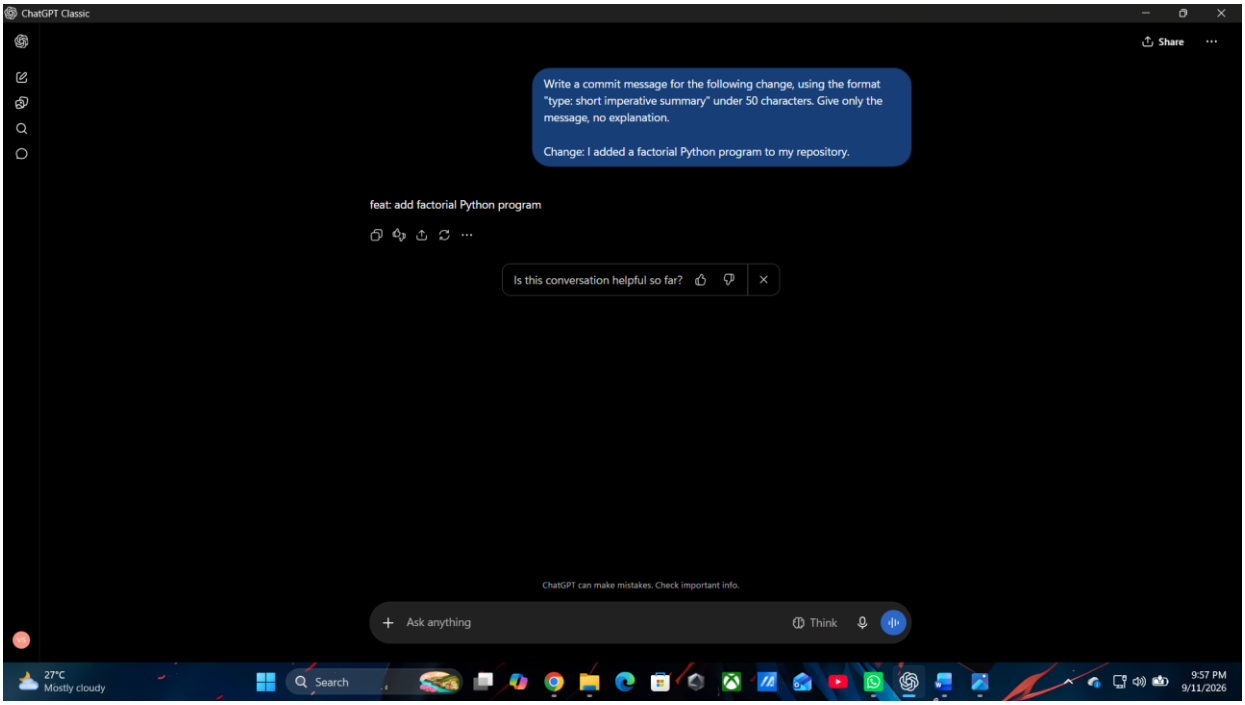


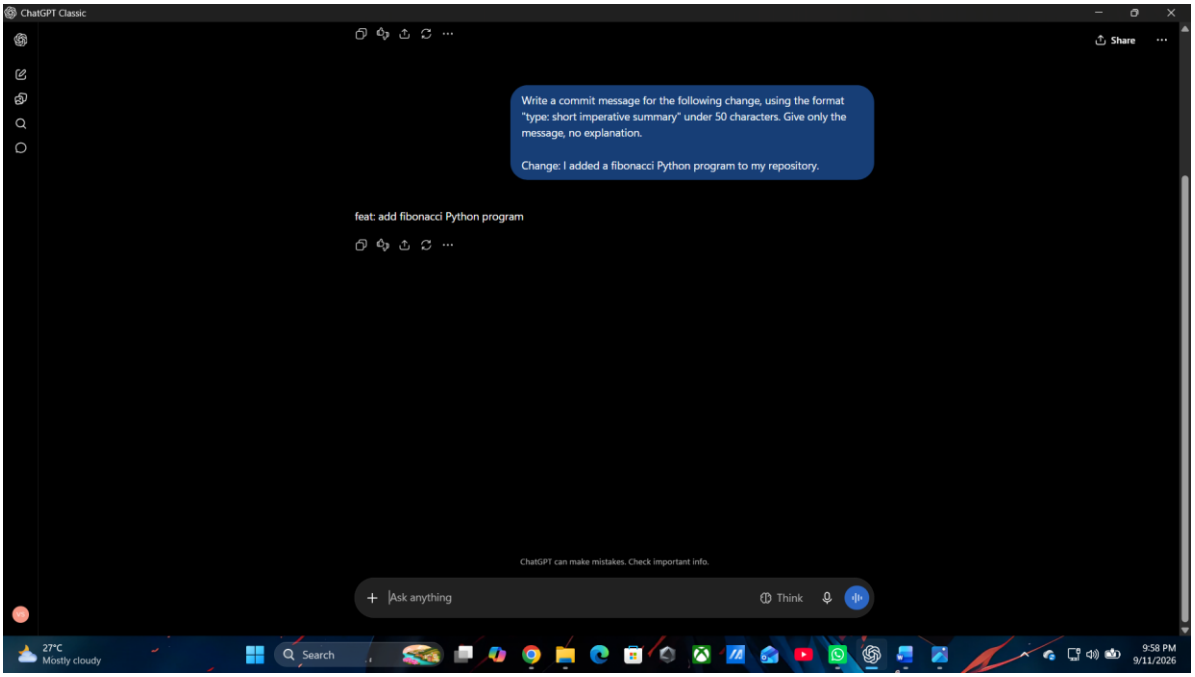
## AI Commit-Message Prompt

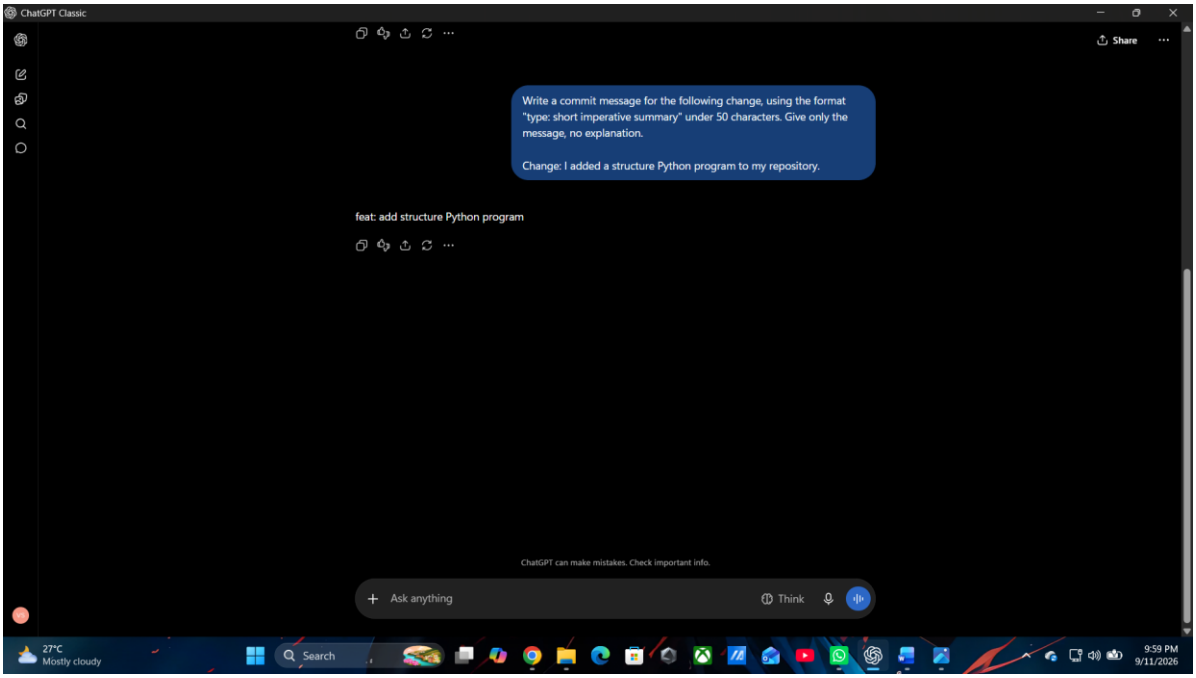
```
Write a commit message for the following change, using the format "type: short imperative summary" under 50 characters. Give only the message, no explanation.
```

## Commit Comparison

| Commit | My message | AI message | Which is clearer, and why? |
|---|---|---|---|
| 1 | feat: add factorial program | feat: add factorial progra | Both are equally clear because they describe the addition of the factorial program directly. |
| 2 | feat: add fibonacci program | feat: add fibonacci program | Both are equally clear because the message immediately tells what program was added. |
| 3 | feat: add student structure | feat: add student structure | Both are equally clear because the message identifies the feature added. |




ChatGPT Classic
Share
Write a commit message for the following change, using the format "type: short imperative summary" under 50 characters. Give only the message, no explanation.
Change: I added a fibonacci Python program to my repository.
feat: add fibonacci Python program
ChatGPT can make mistakes. Check important info.
Ask anything
Think
27°C
Mostly cloudy
Search
9:58 PM
9/11/2026


ChatGPT Classic
Share
Write a commit message for the following change, using the format "type: short imperative summary" under 50 characters. Give only the message, no explanation.
Change: I added a structure Python program to my repository.
feat: add structure Python program
ChatGPT can make mistakes. Check important info.
Ask anything
Think
27°C
Mostly cloudy
Search
9:59 PM
9/11/2026

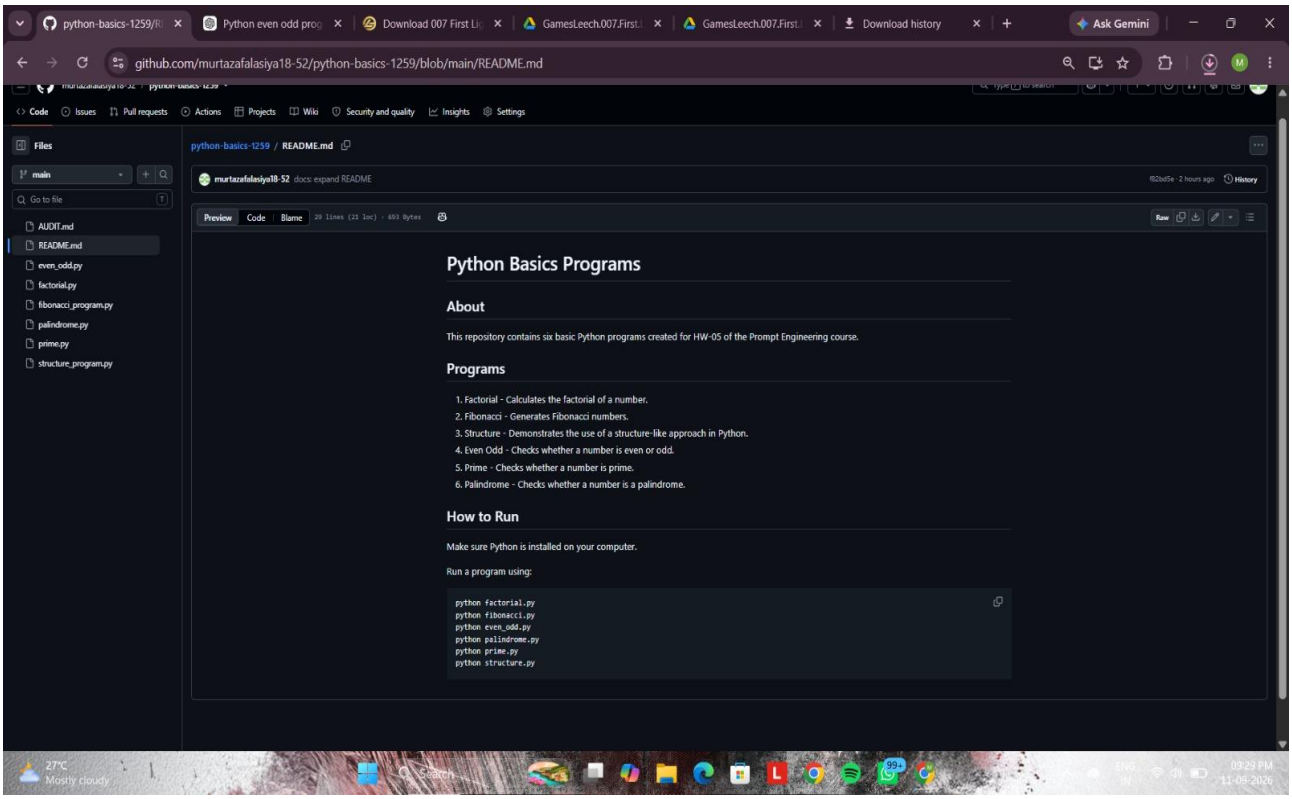
# 3. README: Six Programs + Claim Audit

## README Content

- factorial.py — calculates the factorial of a number using a loop.
- fibonacci.py — generates a Fibonacci sequence for a requested number of terms.
- structure.py — uses a Python class to store and display student information.
- even_odd.py — determines whether an input number is even or odd.
- prime.py — checks whether an input number is prime.
- palindrome.py — checks whether an input value reads the same forward and backward.

## How to Run

```
python factorial.py
python fibonacci.py
python structure.py
python even_odd.py
python prime.py
python palindrome.py
```



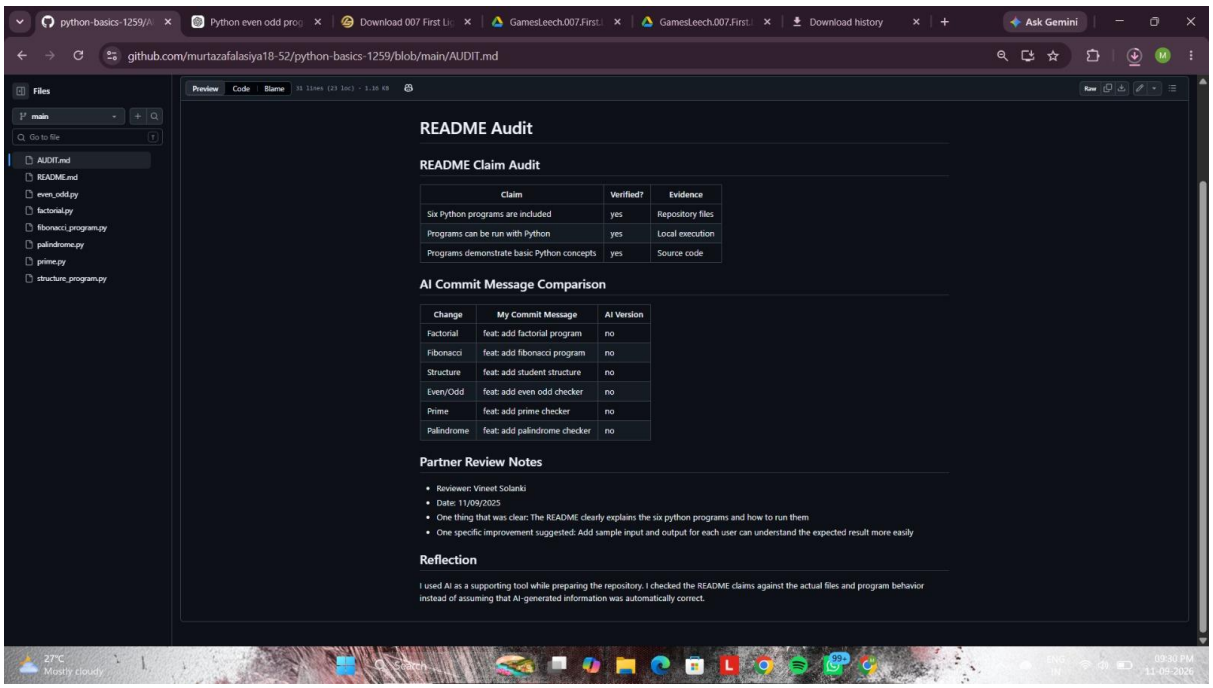
## README Audit Table

| Claim made in README | True? | Evidence / correction made |
|---|---|---|
| Six programs are present. | TRUE | Verified that factorial.py, fibonacci.py, structure.py, even_odd.py, prime.py, and palindrome.py are present in the GitHub repository. |
| factorial.py calculates factorial. | TRUE | Ran the program with a known input and compared the output with the expected factorial result. |
| fibonacci.py generates requested terms. | TRUE | Ran the program with a known input and verified the generated Fibonacci sequence. |
| structure.py stores student information. | TRUE | Checked the code and output to verify that student information is stored/displayed. |
| No external packages are required. | TRUE | Checked the imports in all six Python programs; no external packages are used. |
| All README commands work. | TRUE | Ran the commands listed in the README and verified that they execute successfully. |

# 4. README Audit + Strong Public README Benchmark

## Audit Mindset

The repository is the source of truth. For every README claim, check whether the named file exists, whether described behavior matches the code, whether commands work, and whether dependency or feature claims are supported.



### README Audit

#### README Claim Audit

| Claim | Verified? | Evidence |
|---|---|---|
| Six Python programs are included | yes | Repository files |
| Programs can be run with Python | yes | Local execution |
| Programs demonstrate basic Python concepts | yes | Source code |

#### AI Commit Message Comparison

| Change | My Commit Message | AI Version |
|---|---|---|
| Factorial | feat: add factorial program | no |
| Fibonacci | feat: add fibonacci program | no |
| Structure | feat: add student structure | no |
| Even/Odd | feat: add even odd checker | no |
| Prime | feat: add prime checker | no |
| Palindrome | feat: add palindrome checker | no |

#### Partner Review Notes

- Reviewer: Vineet Solanki
- Date: 11/09/2025
- One thing that was clear: The README clearly explains the six python programs and how to run them
- One specific improvement suggested: Add sample input and output for each user can understand the expected result more easily

#### Reflection

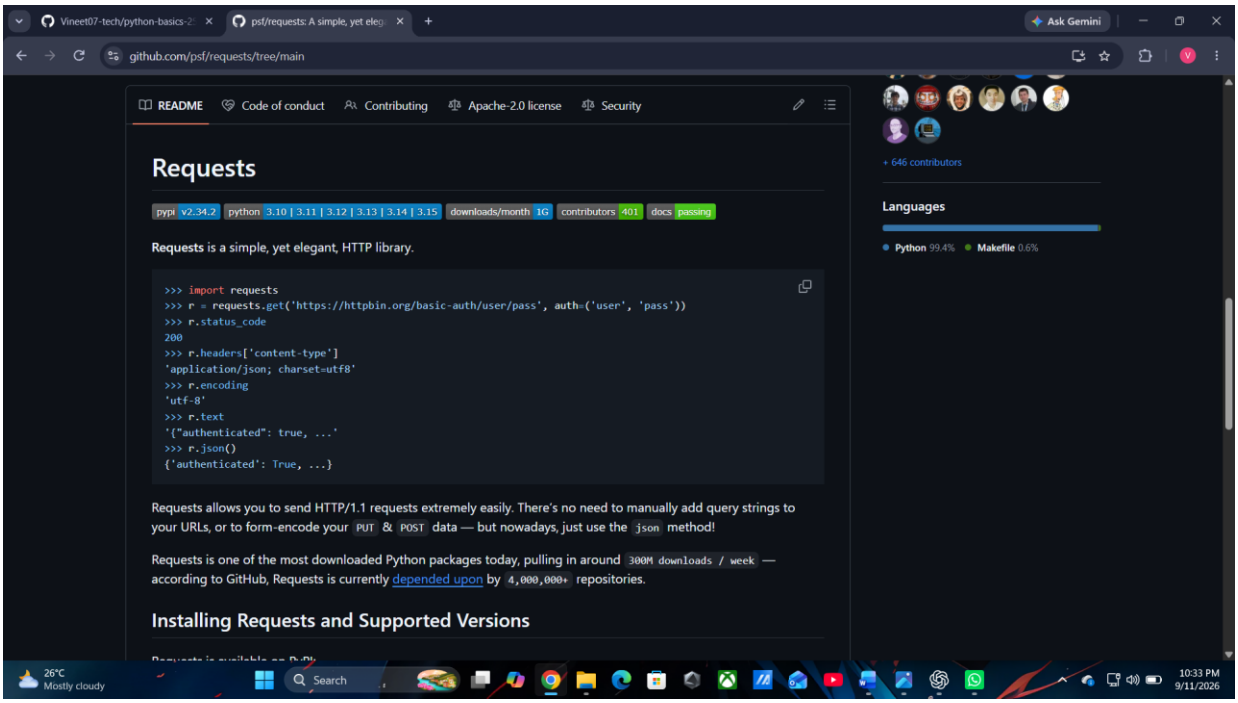
I used AI as a supporting tool while preparing the repository. I checked the README claims against the actual files and program behavior instead of assuming that AI-generated information was automatically correct.

## Benchmark: psf/requests README

The public Requests README is a useful benchmark because it gives a concise project description, an installation command, supported Python versions, feature information, examples, and source/repository information. These qualities can be applied to a smaller student repository.

Benchmark source: GitHub – psf/requests README: https://github.com/psf/requests/blob/main/README.md

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README | Code of conduct | Contributing | Apache-2.0 license | Security

### Requests

pypi v2.34.2 | python 3.10 | 3.11 | 3.12 | 3.13 | 3.14 | 3.15 | downloads/month 1G | contributors 401 | docs passing

**Requests** is a simple, yet elegant, HTTP library.

```
>>> import requests
>>> r = requests.get('https://httpbin.org/basic-auth/user/pass', auth=('user', 'pass'))
>>> r.status_code
200
>>> r.headers['content-type']
'application/json; charset=utf8'
>>> r.encoding
'utf-8'
>>> r.text
'{"authenticated": true, ...'
>>> r.json()
{'authenticated': True, ...}
```

Requests allows you to send HTTP/1.1 requests extremely easily. There's no need to manually add query strings to your URLs, or to form-encode your `PUT` & `POST` data — but nowadays, just use the `json` method!

Requests is one of the most downloaded Python packages today, pulling in around `300M downloads / week` — according to GitHub, Requests is currently depended upon by `4,000,000+` repositories.

### Installing Requests and Supported Versions

+ 646 contributors

Languages

Python 99.4% Makefile 0.6%

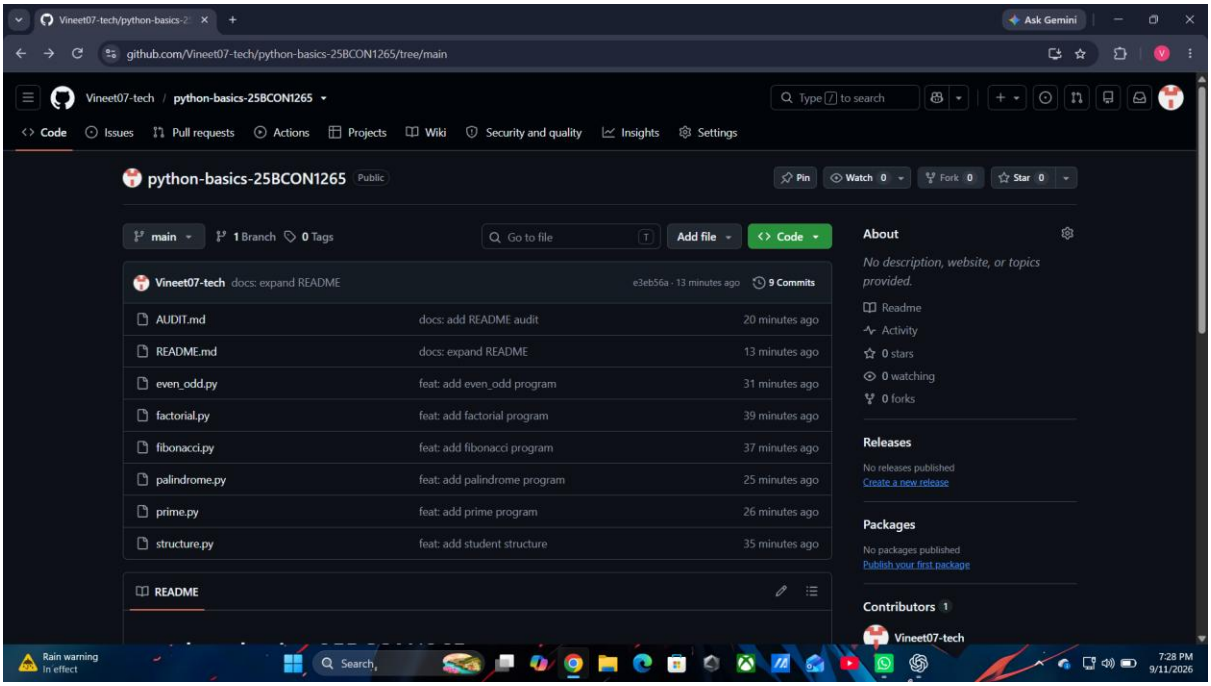
# 5. Peer Review + Personal Reflection

## Peer Repository Review

The peer review requires exchanging public repository links, understanding the repository quickly, verifying two claims, reading commit messages, and giving one specific fix. The partner's notes should be appended to AUDIT.md.

Partner name: Vineet Solanki

Partner repository link: https://github.com/Vineet07-tech/python-basics-25BCON1265




| Review item | Actual notes |
|---|---|
| Claim 1 verified | The repository contains all six Python programs listed in the README. |
| Claim 2 verified | The README commands match the Python filenames in the repository. |
| Commit history observation | The programs were added through separate commits with clear and concise commit messages. |
| One specific fix | Add sample input and output for each program to make the README easier to understand. |

## Personal Reflection

Through this assignment, I learned that a public repository is not only a place to store code but also a record of how the code was created and changed. Clear commit messages make the history easier for another person to understand.

I also learned that an AI-generated README cannot be trusted simply because it is fluent. Statements about files, dependencies, features, and commands must be checked against the actual repository. Unsupported claims should be corrected or removed.

My main lesson is that public documentation is part of professional evidence. Before publishing a README or commit message, I should be able to defend every statement in it.